

Analyzing Twitter Follower Homogeneity and Interactions among Diagnosed and Non-Diagnosed Users with Mental Health Conditions

Tanjima Nasreen Jenia

Introduction:

The prevalence of mental health conditions is a growing public health concern, with millions of individuals affected worldwide. Social networking sites (SNS) have become an essential platform for individuals to connect, share information, and seek peer support related to their mental health. However, little is known about the interactions and homogeneity patterns among Twitter users with self-reported mental health conditions and those without.

This research project addresses this gap by investigating potential differences between the two groups' social support, information-sharing, and engagement behaviors. By exploring the social network structures and interactions of Twitter users with different mental health statuses, the study provides insights into how mental health conditions may affect individuals' online social behaviors and support systems. The findings of this research can help to inform the development of targeted interventions to improve the mental health outcomes of social media users.

Several studies have explored the behavior and interactions between Twitter users. Shoeibi et al. (2022) measured the similarity between Twitter profiles based on the behavioral habits of users. Rusche (2022) studied the homogeneity of followers of politicians from different parties. Durazzi et al. (2021) identified several stable clusters of Twitter users linked to international science and health groups, national elites, and political actors.

Inspired by these studies, this project explores and compares interactions and homogeneity patterns among Twitter users with and without self-reported mental health conditions.

Related Work:

Several studies have analyzed different aspects of Twitter user behavior and network structure. For instance, Shoeibi et al. (2022) proposed a hybrid model that measures the similarity between Twitter profiles using features such as timelines, audience networks, and tweet contents. The study found that the model could detect similarities between profiles with an accuracy of 0.97. Similarly, Rusche (2022) proposed a Homogeneity of Followers metric to calculate the homogeneity of followers of politicians from different parties. The study constructed a network of followers for each politician and party and used the metric to analyze the network structure.

Durazzi et al. (2021) characterized the retweet network to identify the automatic clustering of users and the evolution of their interaction over time with the pandemic's emergence. The study found several stable clusters of Twitter users (super-communities) linked to international groups involved in science and health topics, national elites, and political actors. The authors used the Louvain algorithm for community detection to identify the largest communities in the network.

Yun et al. (2019) analyzed the congruence between a brand's Twitter account and its followers. The authors extracted Big5 personality traits from the text for brand accounts and followers and used the cosine similarity metric to calculate the similarities. Goel et al. (2013) proposed a framework to discover similar users on Twitter. The framework generates similar-to candidates based on whether they have similar followers and uses cosine similarity to measure the similarity between follower graphs of two users.

Overall, these studies provide insights into various aspects of Twitter user behavior and network structure, which can help us understand the social dynamics of the platform.

Methodology:

In this study, we aim to gain insights into the interaction habits and similarities of Twitter users who self-reported mental health conditions and those who do not. To achieve this goal, we will collect data from Twitter users representing both groups, including their ego-network users. We will then clean the collected data and extract features representing Twitter activity and other habitual characteristics. We aim to understand the distribution of these features in the two groups and identify any potential differences. Additionally, we plan to construct network structures for both groups involving their respective network users and analyze any distinguishing features to understand better how their mental health status may impact their social network structure and interactions on the platform.

Data Collection:

We collected our data in two steps. Step 1 involved collecting data for our user-group dataset, i.e., the diagnosed and non-diagnosed/control users with mental health conditions. Users and their posts were extracted from Twitter using its API¹. The collected tweets were public and posted between September 1st, 2020, and August 31st, 2021. We selected the diagnosed users who self-diagnosed by tweeting, "I was diagnosed," or "I've been diagnosed." We had users from nine diagnosed conditions: Anxiety, Bipolar, Depression, Eating disorders, OCD, Schizophrenia, PTSD, ASD, and ADHD. The control/non-diagnosed users were chosen from a pool of candidate users based on their similarity to diagnosed users as measured by the number of messages they posted.

After collecting the user-group dataset, we collected the network-group dataset. First, we extracted the frequently mentioned usernames in our user-group dataset for each user. Then we decided to collect posts and other information by using Twitter API for the top 5 usernames information for each user in our dataset.

¹ Twitter API <https://developer.twitter.com/en/docs/twitter-api>

Data Preprocessing:

We used SoDa-TAP² (Social Data - Toolkit Analysis Platform) for preprocessing the collected datasets. Each user's information was processed through the tool, extracting features like text sentiment, personality features (Big5), hashtags, handles, and influence metrics. After preprocessing the datasets, we removed features that don't represent the Twitter activity and interactions of the users for further analysis.

We also extracted features similar to our user-group dataset for the network-group dataset. We added unique identifiers for each user in the network-group dataset to map with the users in our user-group dataset, which will be necessary for network analysis.

Feature Extraction:

We derived new features based on the users' Twitter activity to better understand their habitual activity and social influence. We calculated the average number of tweets posted daily, the average number of likes, replies, sentiments, emojis, etc., for each user.

To study and identify the pattern between the two groups for the most active period or most active day, we decided to split the time stamp of each tweet for every user. For the most active period of the day, we divided the time into five categories, Morning (5:00-11:59), Afternoon (12:00-16:59), Evening (17:00-20), Night (21:00-23:59), and Midnight(1:00-4:59). After extracting that, we calculated the most active period for each user.

Similarly, we calculated the day of the week from the date of every tweet from each user and derived the most active day of the week. We also measured the average personality score for each user based on the Big5 personality scores of their tweet. Table 1 illustrates the list of extracted features for our analysis.

² SoDa-TAP <https://www.sodatap.ml/en/latest/>

Table 1: List of New Extracted Features

Extracted Features	
avg_tweets_per_day	Total # of Tweets/Total # of Days
avg_tweet_favorite_count	Total likes/Total # of Tweets per User
avg_tweet_reply_count	Total reply/Total # of Tweets per User
avg_text_sentiment	Total sentiment/Total # of Tweets per User
avg_engagement_rate	Total engagement/Total # of Tweets per User
avg_emojis_count	Total emojis/Total # of Tweets per User
avg_tweet_retweet_count	Total retweet/Total # of Tweets per User
most_active_period	When were most tweets posted during the day? (morning, afternoon, evening, night, midnight)
most_active_day	Which day had the most tweets posted by the user?
user_group	Diagnosed/Non-diagnosed
avg_personality measure for o, c, e, a, n	Total personality measure for each o, c, e, a, n values / Total # of Tweets per User

Data Analysis:

This step involves analyzing and investigating the distribution of features in both groups and their networks. We used visualizations, correlations, and clustering to summarize their main characteristics.

First, we generated various distribution plots by exploring the distribution of features in both groups. We wanted to analyze the distribution of features such as the most active period/time of

the day, the most active day of the week, amount of friends, amount of tweets, friends, or the number of mentions, which will indicate their interaction patterns.

Next, we investigated the correlations between features and the two groups to identify any relationships. Finally, we used clustering techniques to identify subgroups or clusters within the two groups based on the most correlated features.

Network Analysis:

In this step, first, we analyzed the characteristics between the users from two groups and their network-groups. After that, we constructed two networks using Gephi³ and examined the distribution of network features such as network size, density, degree distribution, etc. Through these visualizations, we can compare the patterns of Twitter activity and the network characteristics of both groups.

Experimental Evaluation:

In our experimental evaluation, we analyzed the characteristics of two groups of Twitter users and their network-groups. We collected and cleaned data representing both groups, extracted features related to Twitter activity and other behavioral habits, and constructed networks for each group. We then analyzed the distributions of these features and their relationships through visualizations, correlations, and clustering.

Experimental Setup:

We collected 3246 users for the user-group, having 1703 control and 1543 diagnosed users. We had 17 numerical and 4 categorical features. In the diagnosed group, we had 9 specific mental health condition sub-groups. Fig. 1 shows the type and distribution of the 9 diagnosed sub-groups.

³ Gephi <https://github.com/gephi/gephi>

Distribution of Different Mental Health Diagnosis Among the Users

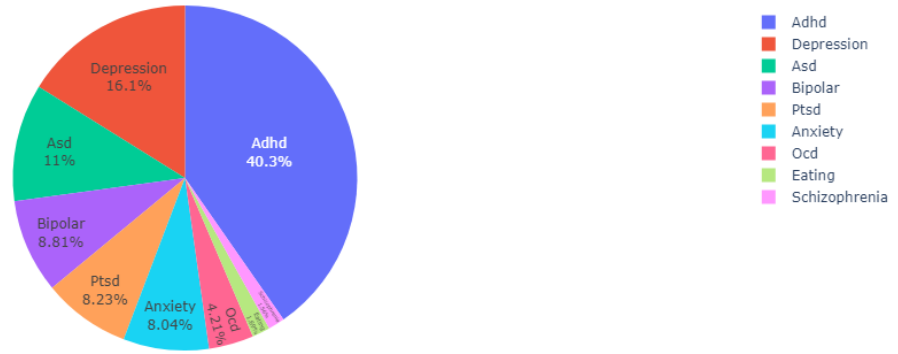


Figure 1. Distribution of Different Diagnosed Sub-groups

Due to time limitations, we created a sub-sample of 300 users from 3246 users for the network-group dataset. We used Stratified sampling for the diagnosed group with a seed of 12 and Random sampling for the control group. For 300 users, we collected 1133 network-users having, 576 from the control and 557 from the diagnosed group.

Results:

Exploratory Data Analysis:

Time of the Day:

From the Fig. 2, we can see that for both user groups, the most active time of the day is midnight (1:00-4:59), followed by afternoon (12:00-16:59). Furthermore, from Fig. 3, we can distinguish two diagnosed groups based on the most active period of the day. People diagnosed with ASD, Anxiety, Bipolar, ADHD, OCD, PTSD, and Eating disorder tend to post tweets mostly at midnight (1:00-4:59), whereas people diagnosed with Depression and Schizophrenia post tweets mainly during the morning (5:00-11:59).

Time period of the day when most tweets are posted among the two groups

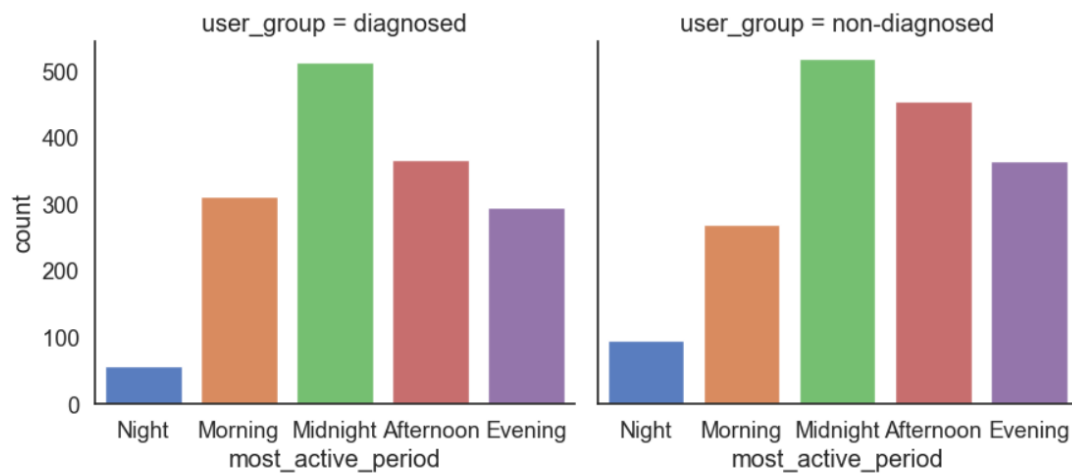


Figure 2: Most Active Time of the Day for the User-group

Tweet posted based on the period of the day among different diagnosed sub-groups

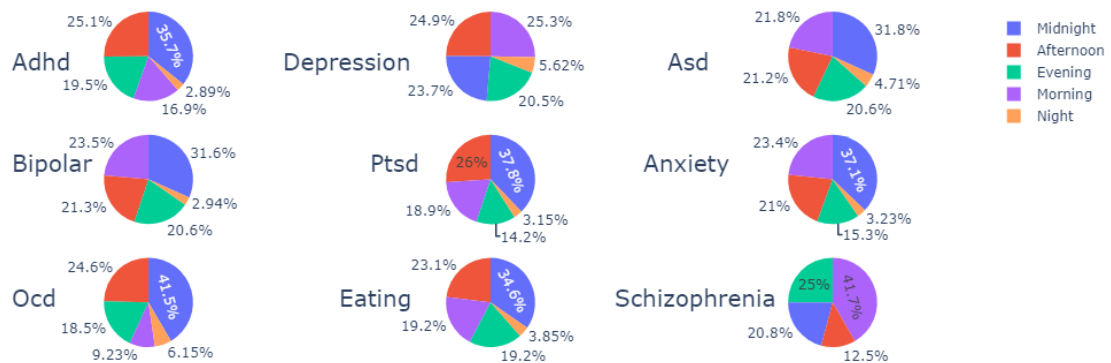


Figure 3: Most Active Time of the Day for the Diagnosed Sub-groups

Day of the Week:

Fig. 4 shows that both user groups are mainly active on weekdays than weekends. The diagnosed group is primarily active on Wednesday, whereas the non-diagnosed/control group is active on Tuesday. Fig. 5 represents the diagnosed sub-groups where we can see that people with Bipolar, OCD, Schizophrenia, ADHD, PTSD, and Anxiety mostly post tweets on Wednesday. People with Depression and ASD stay active on Thursday and Monday, respectively. Conversely, people diagnosed with Eating Disorders mainly stay active over the weekends, particularly on Friday and Sunday.

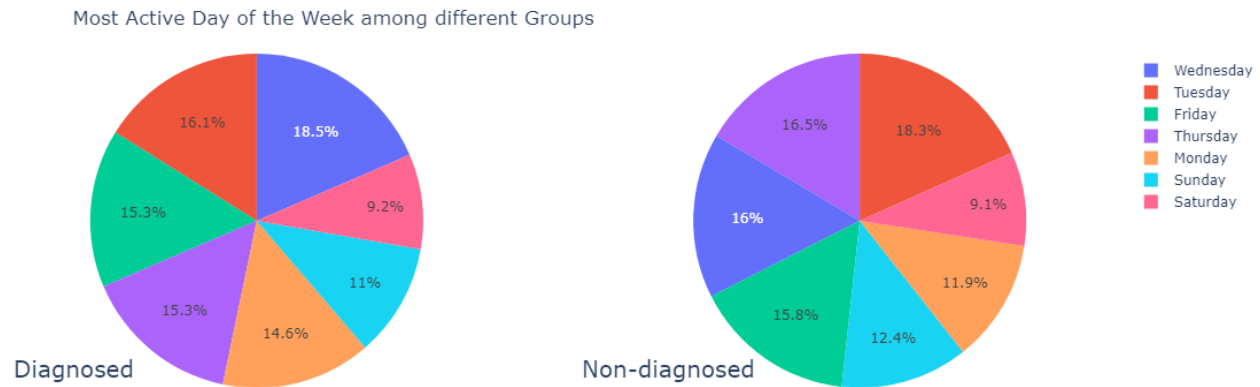


Figure 4: Most Active Day of the Week for the User-groups

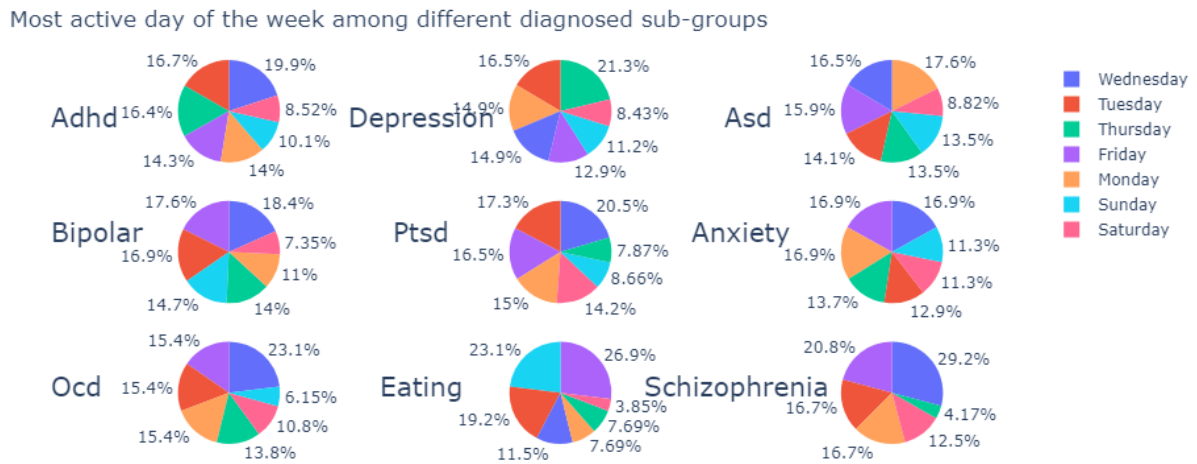


Figure 5: Most Active Day of the Week for the Diagnosed Sub-groups

Tweeting Magnitude:

We can get the number of posted tweets from every user. Since the number of tweets is a continuous variable, we decided to bin them into three categories to understand their distribution among the user groups and the diagnosed sub-groups. Since the distribution of the number of tweets was heavily skewed towards the right, we decided to use the Freedman-Diaconis rule to determine the bin sizes using the interquartile range (IQR).

The three categories were Low (19 - 6921), Medium (6922 - 10686), and High (10687 - 895987). For both user groups, the magnitude of tweets was high. From Fig. 6, we can say that people diagnosed with Eating Disorders, ASD, Depression, and Schizophrenia have a low tweeting tendency. In contrast, people with ADHD, PTSD, Bipolar, OCD, and Anxiety have a high tweeting tendency.

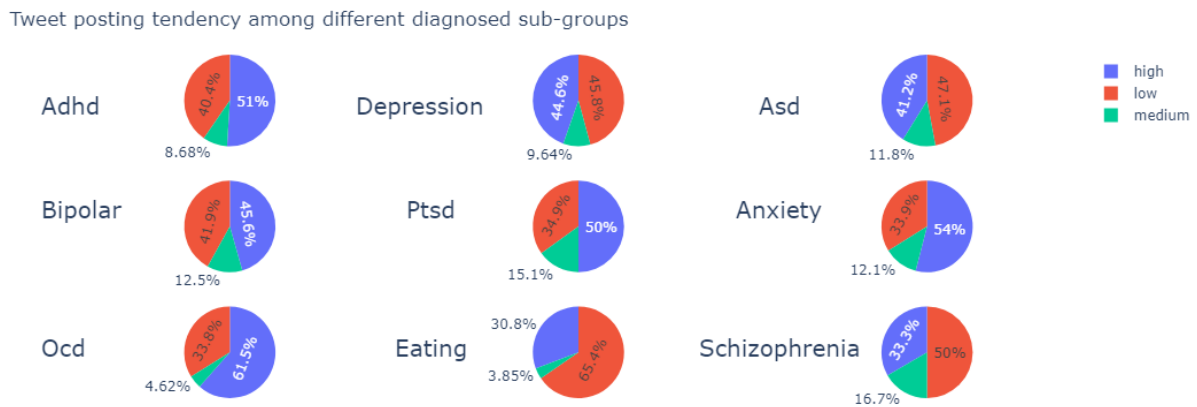


Figure 6: Amount of Tweets among the Different Diagnosed Sub-groups

Mentioning Other Users:

Similarly, to understand the distribution of mentioning or interacting with other users on Twitter among the user groups, we binned them into three categories using the Freedman-Diaconis rule. The range for the three categories are Low (0-149), Medium (150-218), and High (219-3590). Both user groups fell into the High category regarding mentioning other users in their tweets. But in Fig. 7, we can see that people diagnosed with Schizophrenia have a lower tendency to mention other users in their tweets.

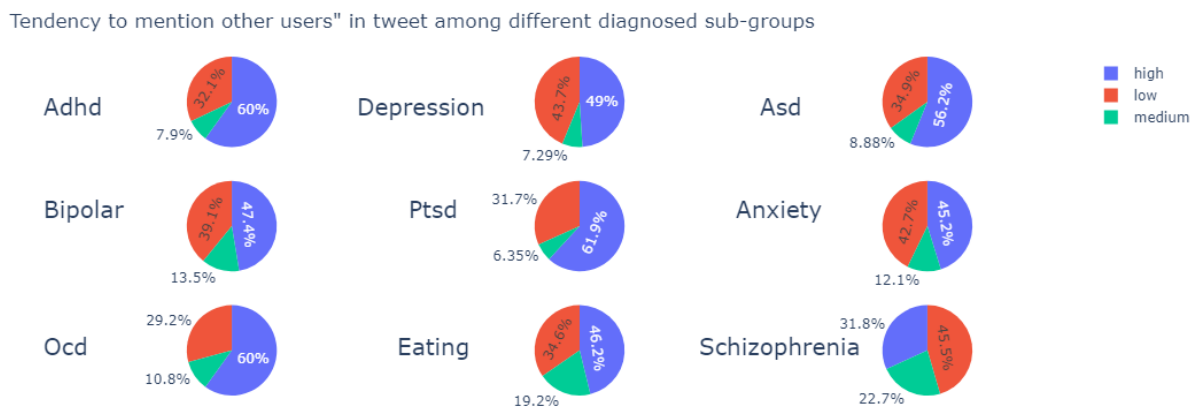


Figure 7: Tendency to Mention Other Users among the Diagnosed Sub-groups

Amount of Friends:

Again, to understand whether mental health-related issues impact the number of friends a user can have, we also binned the number of friends for each user using the previously mentioned approach. We binned it into three categories, Low (0 - 390), Medium (340 - 537), and High (538 - 104314). For both user groups, we noticed a higher number of friends. But in Fig. 8, we noticed that the people diagnosed with Eating Disorders have a lower magnitude of friend-circle.

Magnitude of Friend-circle among different diagnosed sub-groups

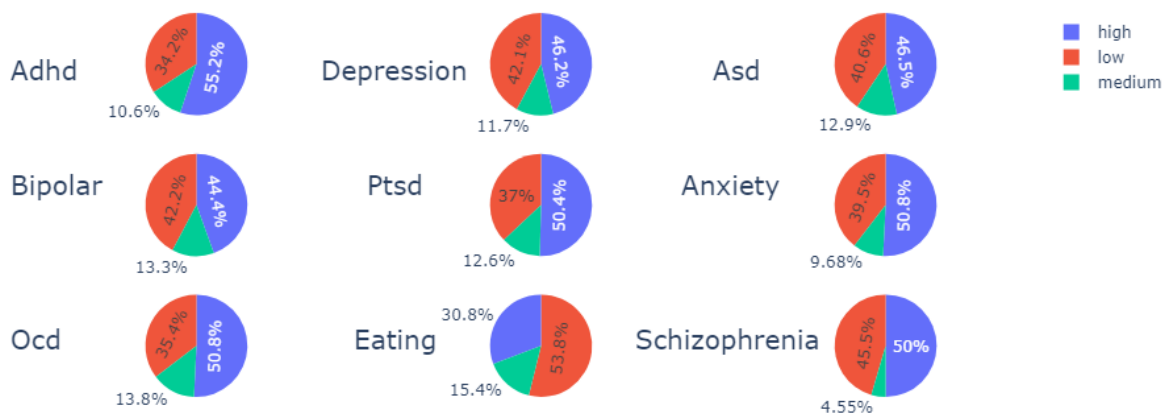


Figure 8: Magnitude of Friend-circle among Different Diagnosed Sub-groups

Tweet Sentiment:

In this section, we wanted to analyze if there exists any relationship between the tweets' sentiment and a user's mental status. For this reason, we also binned the average tweet sentiment of a user into three categories, Negative (-0.511 - 0.0), Neutral (0.0 - 0.291), and Positive (0.291 - 0.6922). Our results show that both user groups mainly had Neutral toned tweets, but the non-diagnosed/control group tended to post more positive tweets. Although, in Fig. 9, we can see that people diagnosed with Eating Disorders, PTSD, and Schizophrenia have comparatively more negative tweets than the other diagnosed groups.

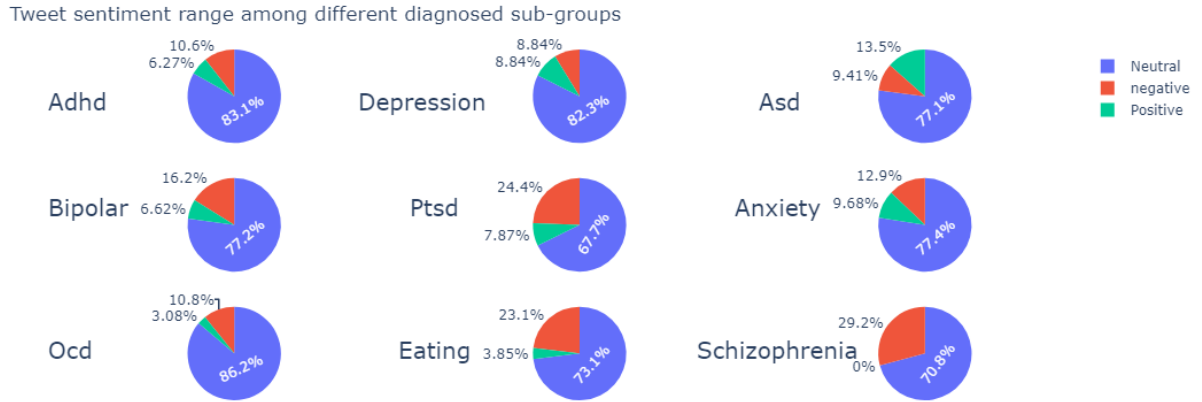


Figure 9: Tweet Sentiment Range among Different Diagnosed Sub-groups

Statistical Analysis:

The user_group feature in our diagnosed and control dataset acts as a target/class label indicating whether a user belongs to a diagnosed group or control. We performed two statistical tests to understand the correlations between the other features and the user_group. We performed a chi-squared test to understand the relationship between the categorical features and user_group and a point-biserial test for the numerical ones.

We implemented the chi-square test with a significance level of 0.050 with a null hypothesis (H0) that there is no association between the variables and the user_group. Table 2 demonstrates the chi-squared test score of most active period, where we can see that the p-value is < 0.05 and the test statistic is greater than the critical value. So, we can reject the null hypothesis and see a statistically significant association between the most active period of the day and the user_group.

Table 2: Chi-squared Test Score

Categorical Features	Score			
	P-value	Critical value	Test Statistic	DOF
Most active period of the day	0.00022	9.488	21.827	4
Most active day of the week	0.0623	12.592	11.984	6
Diagnosis (9 sub-groups, 1 control group)	0.000	16.919	3246.000	9

For the numerical variables, we applied the point-biserial test with the same null hypothesis and $p\text{-value} < 0.05$. In Table 3, we only showed the variables associated with the `user_group`. The table shows that Conscientiousness and Neuroticism are moderately positively correlated with the `user_group` and are statistically significant.

Table 3: Point-Biserial Test Score

Numerical Features	Score	
	P-value	Coefficient
User friends count	0.011	0.044
Avg text sentiment	5.584e-06	0.0796
Openness	0.0046	-0.0496
Conscientiousness	2.013e-17	0.148
Neuroticism	1.286e-33	0.2098

Clustering:

From the results obtained from the statistical analysis, Conscientiousness and Neuroticism have a positive correlation with the two user groups. To further understand their effects on the diagnosed sub-groups, we applied multiple clustering algorithms on the user-group dataset based on the value of Conscientiousness and Neuroticism. We obtained 10 clusters from Agglomerative and KMeans clustering and 3 clusters, and 1 noise cluster from DBSCAN.

From Fig. 10 and 11 for Agglomerative clustering, we can see that clusters 0 and 5 grouped people with ADHD, PTSD, Depression, and Schizophrenia together, whereas clusters 4, 6, 7, and 9 consist of control users. A similar result can also be seen for KMeans in Fig. 14 and 15, where clusters 2 and 7 had the same diagnosed group of ADHD, PTSD, Depression, and Schizophrenia, and clusters 0, 3, 6, and 8 had control users. Again, from DBSCAN in Fig. 12 and 13, we interestingly got two different clusters where cluster 0 grouped ADHD, PTSD, Depression, and Schizophrenia, and cluster 2 solely represented the control group.

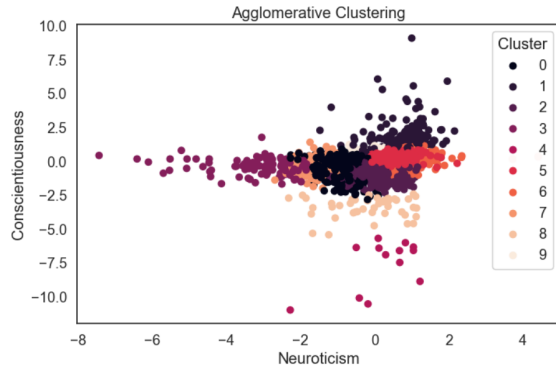


Figure 10: Agglomerative Clustering

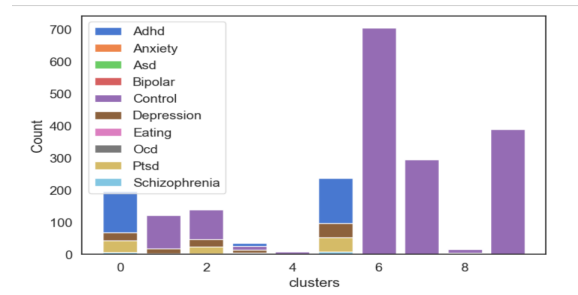


Figure 11: Agglomerative Clustering Groups

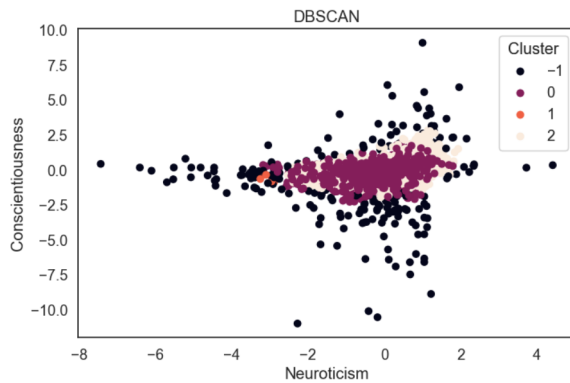


Figure 12: DBSCAN

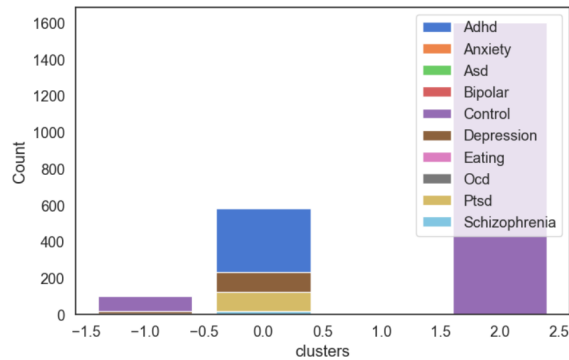


Figure 13: DBSCAN Groups

So we can conclude that people with ADHD, PTSD, Depression, and Schizophrenia have a similar range of values for Conscientiousness and Neuroticism and the majority of the control group have a different range of values for Conscientiousness and Neuroticism from the diagnosed group.

Moreover, we can also see this same diagnosed sub-group consisted of a combination of people diagnosed with ADHD, PTSD, Depression, and Schizophrenia from the results of our Exploratory data analysis: Most Active Time of the Day (ADHD and PTSD, Depression and Schizophrenia), Most Active Day of Week (Schizophrenia, ADHD, and PTSD), Tweeting Magnitude (ADHD and PTSD, Depression and Schizophrenia), Mentioning Other Users

(ADHD, PTSD, and Depression), Amount of Friends (ADHD, PTSD, Depression, and Schizophrenia) and finally, Tweet Sentiment (Schizophrenia and PTSD).

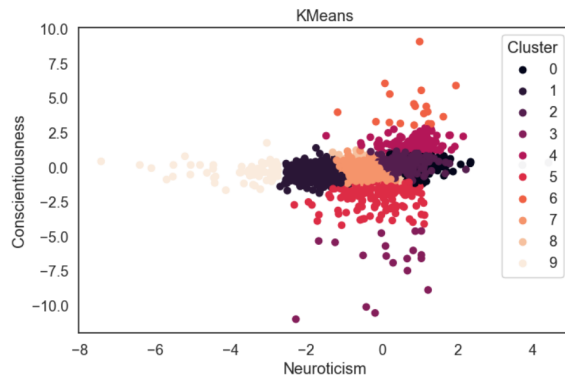


Figure 14: KMeans Clustering

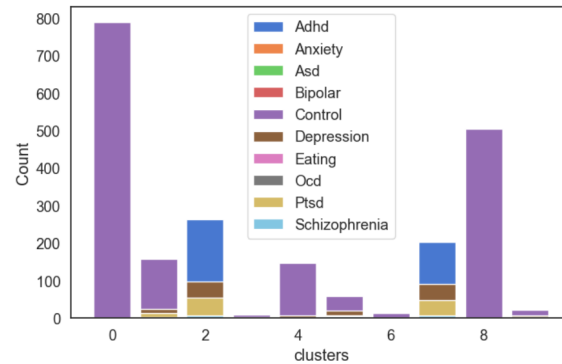


Figure 15: KMeans Clustering Groups

Network Analysis:

This section aims to understand the similarities between diagnosed and control users and their respective network groups. From our above analysis, we could conclude a relationship exists between the mental health diagnosis and the most active period of the day, and the number of friends they have; therefore, we would like to see if this also remains true for the network groups for both sampled diagnosed and non-diagnosed users.

Time of the Day:

We found a higher similarity between the diagnosed users and their network groups by analyzing the distributions of the most active period of the day for the two user-groups and their respective network groups. Midnight was the most active period for the diagnosed users and their networks. But for the control users and their network, it was different. We also wanted to see if more similarities between the diagnosed sub-groups and their networks could be found.

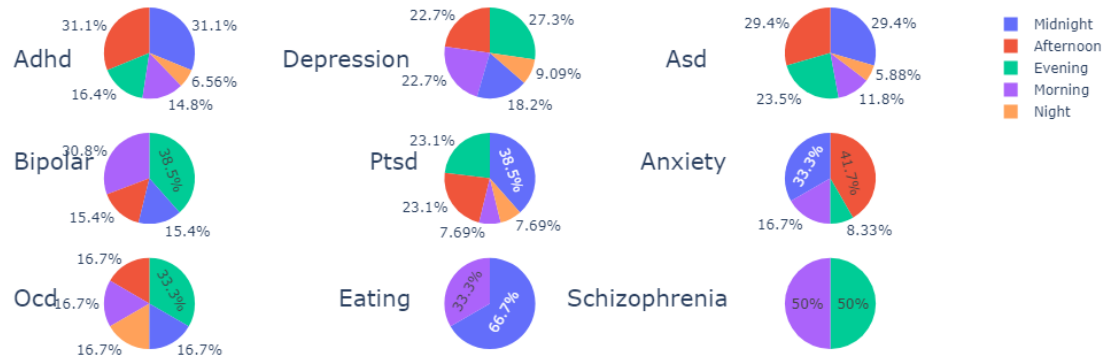


Figure 16: Most Active Period of the Day for Diagnosed Sub-groups

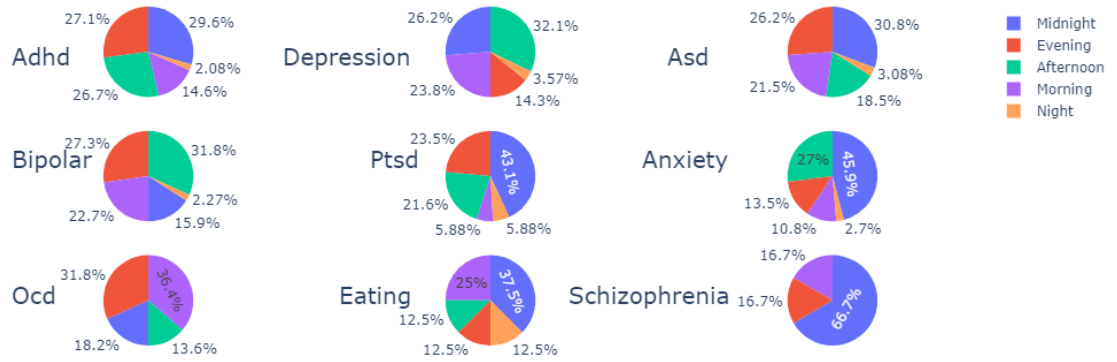


Figure 17: Most Active Period of the Day for Different Diagnosed Network-groups

Fig. 16 and 17 represent the diagnosed sub-groups and their respective network groups. We see similarities between multiple mental health condition groups and their networks. We can infer that for PTSD users and their network, the most active periods are evening and midnight. For Anxiety, it is afternoon and midnight. Users diagnosed with Bipolar and Ocd and their respective networks have the evening in common as the most active period of the day. For the Eating Disorders users and their network, that are morning and midnight. The ADHD users and their work have similar active periods from the afternoon to midnight.

Amount of Friends:

From Fig. 18 and 19, we can see that people diagnosed with Depression, PTSD, Anxiety, and Eating Disorders and their respective network groups have a low amount of friends. Conversely, people diagnosed with ADHD, Bipolar and their network groups have a high amount of friends.

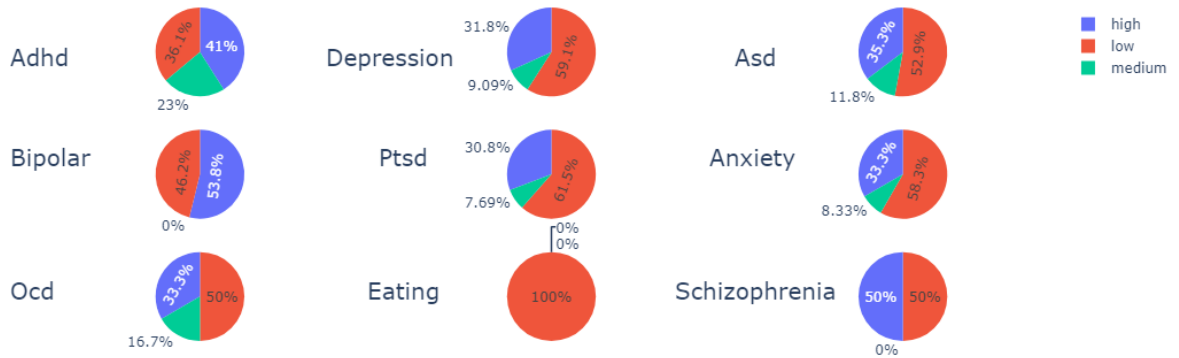


Figure 18: Magnitude of Friend-circle among Different Diagnosed Sub-groups



Figure 19: Magnitude of Friend-circle among Different Diagnosed Network-groups

Network Structure:

To build and analyze the network structure for both the control and diagnosed groups, we first extracted the nodes and edges for both groups. Nodes were the individual users we got from both the user-group dataset and network-group dataset. The nodes were labeled by the diagnosed sub-group/control they belonged to or part of the network. The edges represented the connection between users from the user-group and their network-group users. The weights for each edge were the number of times our diagnosed/control user mentioned that particular network-user.

Apart from the 9 diagnosed groups and 1 control group, we added a new group (Mixed) for the network users mentioned by multiple diagnosed and non-diagnosed groups. Fig. 20 and 21 illustrate the control and diagnosed network structure. Here, each node's size is proportional to the number of degrees. And the thickness of the edges is proportional to the mention frequency.

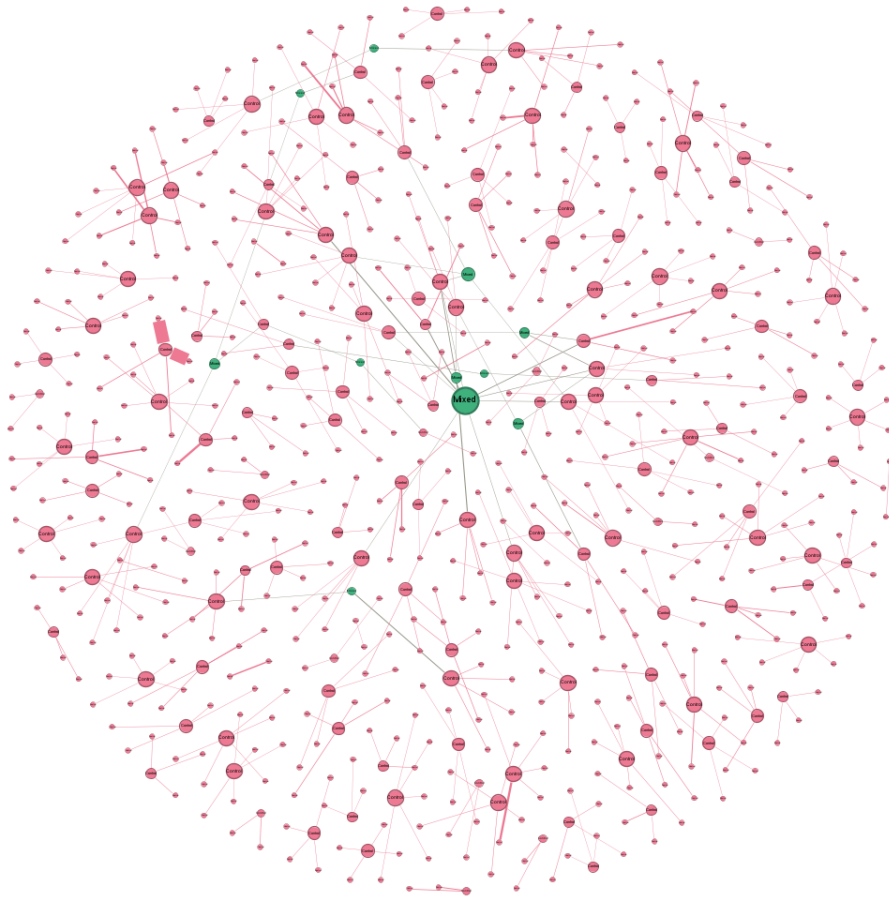


Figure 20: Control Network Structure

From Fig. 20 and 21, we can see that the thicker edges are more prevalent in the diagnosed network compared to the control network. To better understand the differences between the two networks, we measured various metrics, as shown in Table 4.

Table 4: Network Metrics

	Diameter	Radius	Avg Path Length	Modularity	# of Communities	Avg Degree
Control	8	1	3.2	0.928	132	1.64
Diagnosed	4	1	1.7	0.969	143	1.59

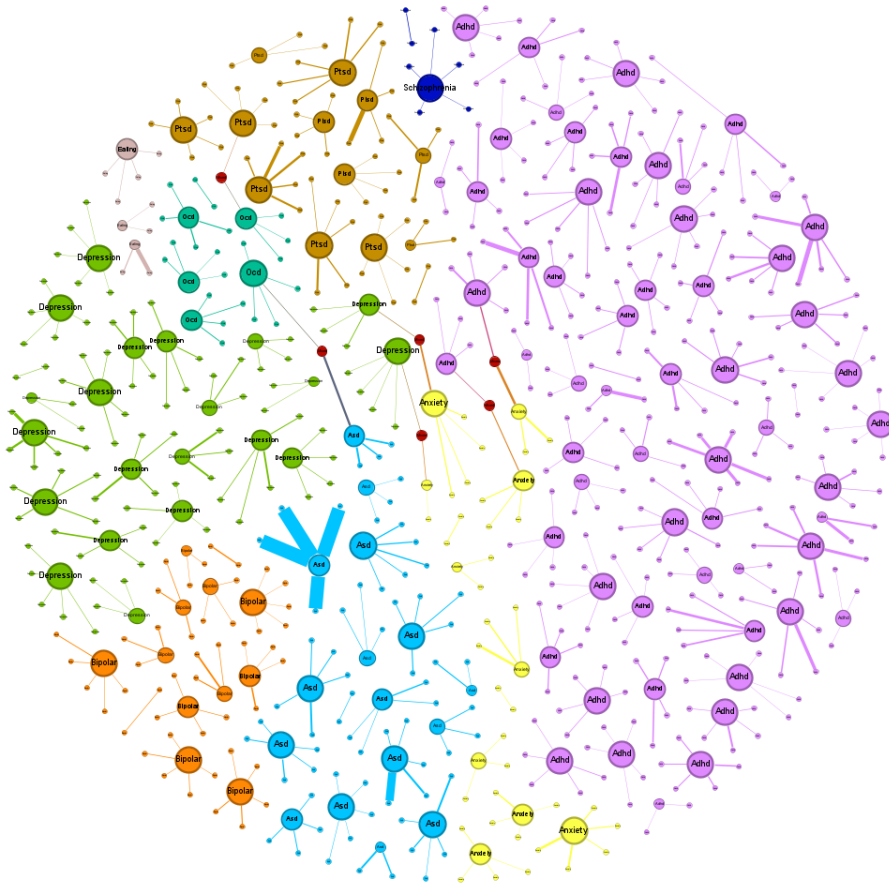


Figure 21: Diagnosed Network Structure

From Table 4, we can infer that the diagnosed network is more densely connected than the control network. It has a smaller diameter of 4 and an average path length of 1.7, indicating the average distance between any two nodes. Again, the modularity and number of communities of the diagnosed network are higher than the control network. It suggests that the diagnosed network has more densely connected communities or modules, which are more distinct than the control network.

Conclusion:

This project aimed to investigate the potential differences between the diagnosed and non-diagnosed groups' social support, information-sharing, and engagement behaviors on social media. It also investigated the interactions and homogeneity patterns among those two groups

with their respective network users. Our analysis showed that the Twitter activities between the two groups are similar in most cases, but the interaction and engagement habits between the diagnosed sub-groups differ. Also, we can see distinct differences between personality traits and tweet sentiment between the diagnosed and non-diagnosed groups.

Moreover, the network analysis between the diagnosed and control group and their network users showed promising results. From there, we can assume that the diagnosed users hold more similarities with their respective network users. We can also notice that the diagnosed network is more tightly connected and has a diverse structure than the control network. It is worth mentioning that the project's most challenging aspect was accumulating and preprocessing the users' and their networks' information and generating scalable insights. Since the network analysis was done on a subset of our actual population dataset, in the future, we aim to expand the analysis to a large-scale dataset that will more accurately represent the real-world population.

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